higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING TECHNOLOGY
NQF LEVEL 2

1 MARCH 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1	1.1.1	B		
	1.1.2	C		
	1.1.3	D		
	1.1.4	D		
	1.1.5	C		
			(5 × 1)	(5)
1.2	1.2.1	B		
	1.2.2	C		
	1.2.3	D		
	1.2.4	A		
	1.2.5	F		
			(5 × 1)	(5)
1.3	1.3.1	False		
	1.3.2	True		
	1.3.3	False		
	1.3.4	False		
	1.3.5	True		
			(5 × 1)	(5)
1.4	1.4.1	Kickback		
	1.4.2	Chuck		
	1.4.3	Power		
	1.4.4	Pulley		
	1.4.5	Compression		
			(5 × 1)	(5)
				[20]

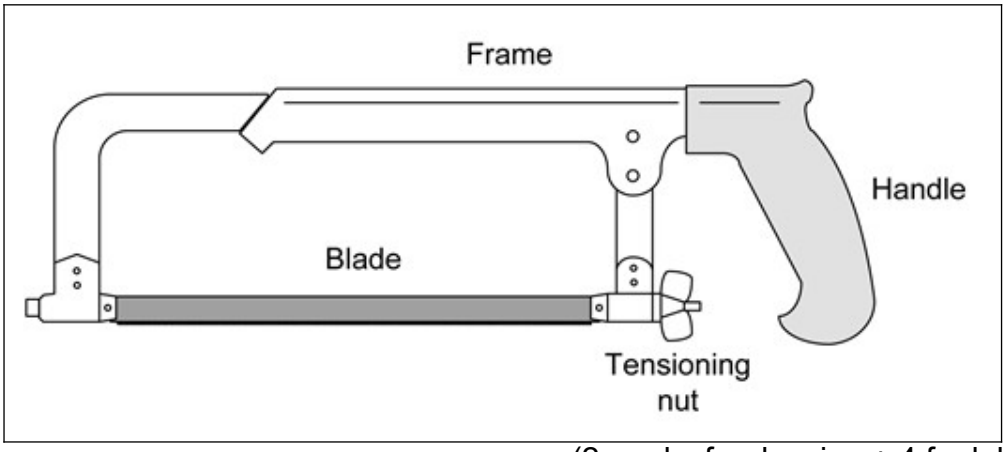
QUESTION 2

2.1
A – Claw hammer
B – Side-cutter plier
C – Allen key
D – Double open-ended spanner
E – Longnose plier
F – Star/Phillips screwdriver (6 × 1) (6)

2.2
- Incorrect and careless use
- Blade too soft
- Blade too hard
- Wrong blade type for type of metal (4)

2.3 Feeler gauge (1)

2.4



(2 marks for drawing + 4 for labels) (6)
[17]

QUESTION 3

3.1 3.1.1 Hydraulic press (1)

3.1.2 A – Pressure gauge
B – Pump handle/lever
C – Ram
D – Adjustable table (4 × 1) (4)

3.2
- Kind of abrasive
- Grain size
- Bonding material
- Grade
- Structure (5)

3.3 3.3.1
- Use the correct key to tighten the chuck.
- Secure the job before drilling.

3.3.2
- Wear safety glasses to prevent eye injuries.
- Never stop the drill spindle by hand.

(2 × 2) (4)
[14]

QUESTION 4

4.1
- Working gauges
- Inspection gauges
- Master gauges (3)

4.2
- Working gauges are used at the machine or production line producing components that must be measured.
- Inspection gauges are more accurate than working gauges and are used by inspectors on a batch of mass-produced components.
- Master gauges are very accurate and are used to check that working and inspection gauges are accurate. (3)

4.3 4.3.1 Callipers are used for taking transfer measurements or to compare external and internal measurements.

4.3.2 A straight edge is used to compare the straightness of the surface of a workpiece to the straightness of the straight edge.
(2 × 2) (4)

4.4
- Keep the thread gauge free of dust and sand.
- Oil it lightly to prevent rusting.
- Slide blades into protection holder after use. (3)

[13]

QUESTION 5

5.1 5.1.1 Outside micrometre (1)

5.1.2 A – Anvil
B – Spindle
C – Barrel/Sleeve
D – Ratchet (4 × 1) (4)

5.2
- Turn the spindle fully onto the anvil ensuring there is no dirt between.
- The datum line on the barrel must correspond to the zero on the thimble graduation.
- If they do not correspond, engage the spindle lock and using the adjusting spanner; adjust the barrel so that the datum line corresponds with zero on thimble. (3)

[8]

QUESTION 6

6.1 W = m × g✓
= 68 × 9,81✓
= 667,08 N✓ (3)

6.2 6.2.1 K
6.2.2 A
6.2.3 mol
(3 × 1) (3)

6.3 6.3.1 0,9 kg
6.3.2 3,5 ℓ
(2 × 1) (2)

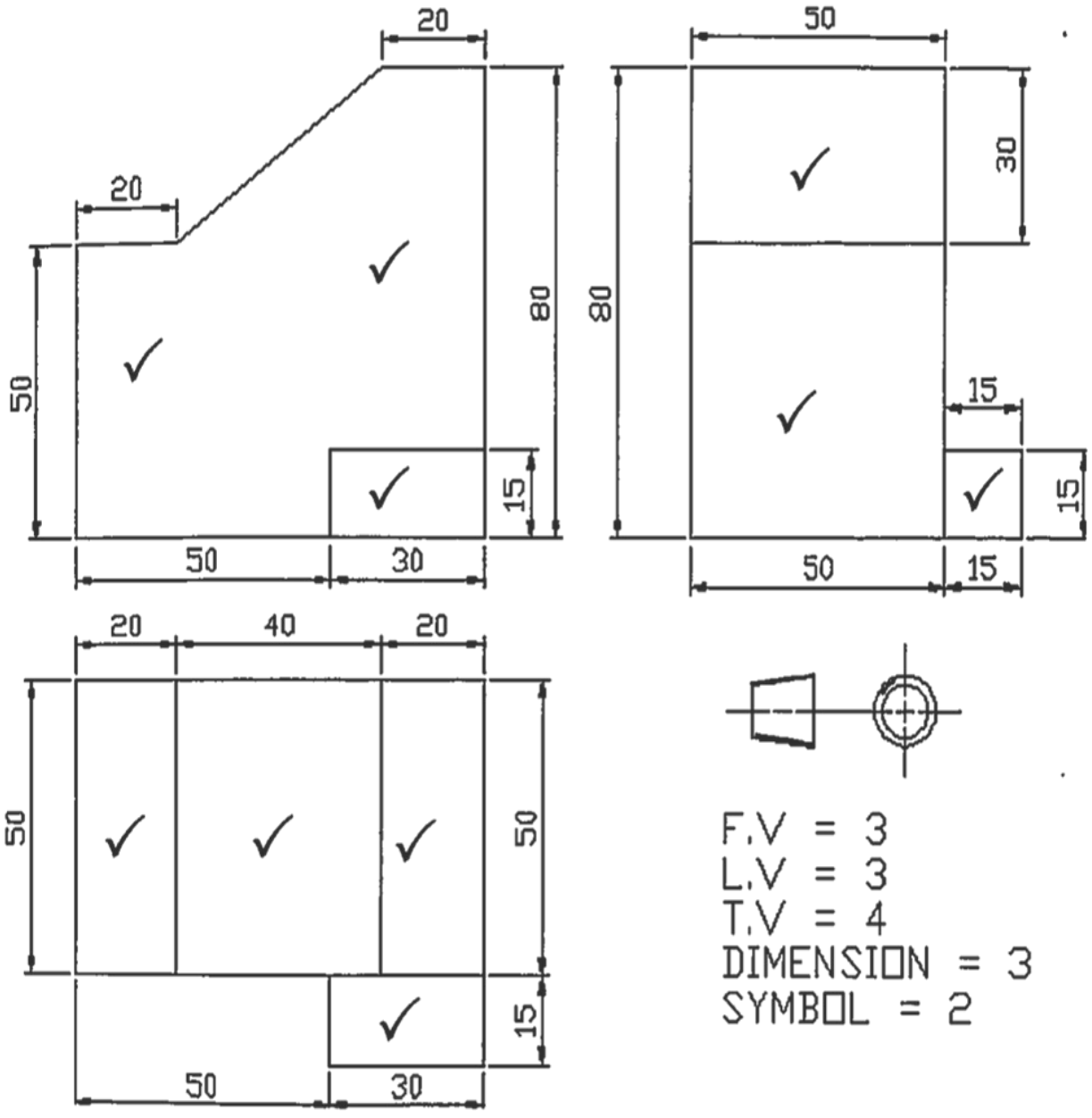
6.4 6.4.1 Jenny calliper/Odd-leg calliper (1)

6.4.2 A – Scriber
B – Odd leg (2 × 1) (2)

6.4.3
- Used to find the centre of a round workpiece
- Used to scribe parallel lines to a reference face (2)

[13]

QUESTION 7



[15]

TOTAL: 100

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